



The potential of Plant Growth Promoting Rhizobacteria for farming

Plant Growth Promoting Rhizobacteria:

- (free living) soil bacteria colonizing plant roots
- stimulate plant growth
- control pathogens

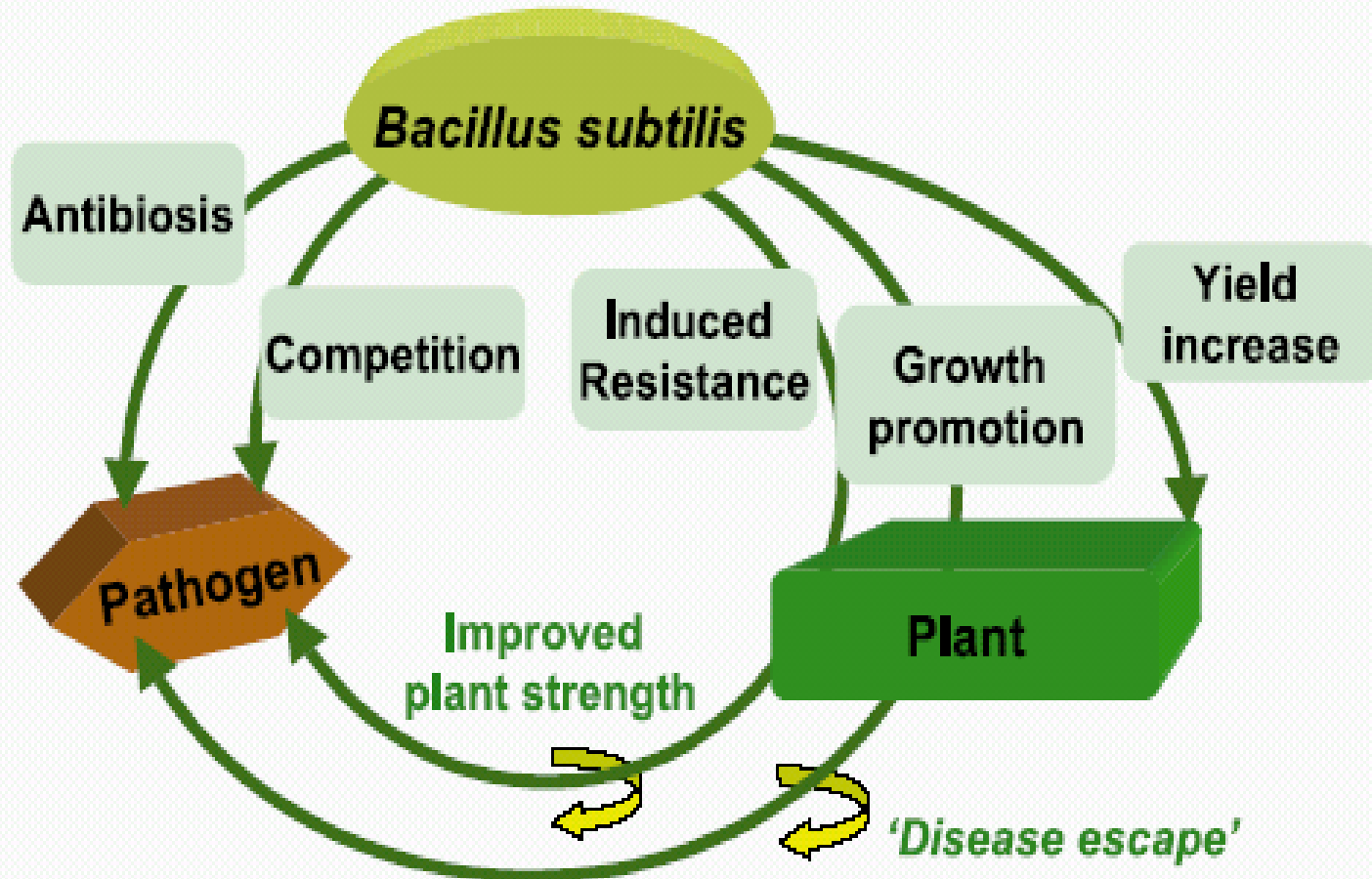
- *Bacillus* - producing spores
- *Actinomyces* - producing spores
- *Azotobacter* - fix nitrogen
- *Azospirillum* - endophytic bacteria / fix nitrogen
- *Pseudomonas* - root biofilm

- *Acetobacter*, *Burkholderia*, *Enterobacter*, *Serratia*, *Paenibacillus* spp.

Mechanisms:

- phytohormones
- solubilizing of P
- nitrogen fixation
- antagonistic effects
 - direct competition for 'rhizosphere space'
 - production of antibiotics
 - siderophores
 - lytic enzymes
- induction of systematic resistance

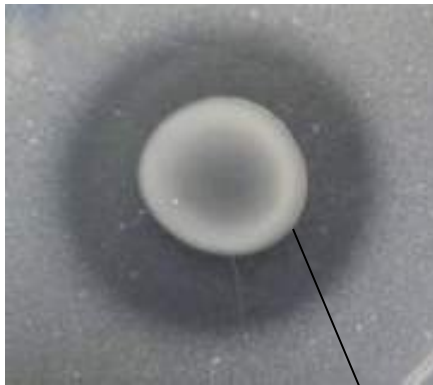
INTERACTION PGPR - PLANT



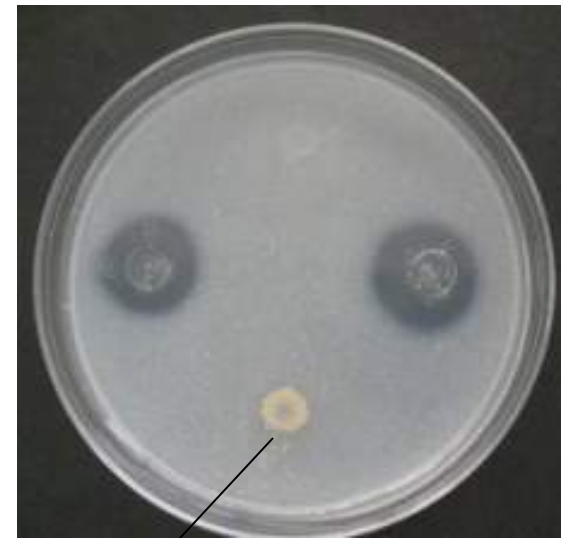
Phosphate Solubilization

Using growth medium with Tricalcium phosphate (NBRIP).

Bacterial mechanisms: phosphatase enzymes or production of acids



Solubilization of P



No solubilization



Pure mineral patches of either potassium feldspar (K) or quartz (Q), introduced at the mycelial front in flatbed peat containing *Pinus sylvestris* seedlings colonized by *Hebeloma crustuliniforme*. Fifteen weeks after introduction, fungal proliferation was more intense in patches of K-feldspar compared to patches of quartz. From Rosling et al. (2004).

Factors Influencing Nutrient Availability

- **Presence of nitrogen fixers**
- **Microbial activity**
- **Fire**
- **Precipitation patterns**
- **Soil drainage**
- **Soil temperature, moisture**

Classification of Soils Based on the Functions of Microorganisms



Disease-Inducing Soils. In this type of soil, plant pathogenic microorganisms such as *Fusarium* fungi can comprise 5 to 20 percent of the total microflora.

If fresh organic matter with a high nitrogen content is applied to such a soil, incompletely oxidized products can arise that are malodorous and toxic to growing plants. Such soils tend to cause frequent infestations of disease organisms, and harmful insects. Thus, the application of fresh organic matter to these soils is often



Probably more than 90 percent of the agricultural land devoted to crop production worldwide can be classified as having disease-inducing soil. Such soils generally have poor physical properties, and large amounts of energy are lost as "greenhouse" gases, particularly in the case of rice fields. Plant nutrients are also subject to immobilization into unavailable forms.



Disease-Suppressive Soils.

- The microflora of disease-suppressive soils is usually dominated by antagonistic microorganisms that produce copious amounts of antibiotics. These include fungi of the genera *Penicillium*, *Trichoderma*, and *Aspergillus*, and actinomycetes of the genus *Streptomyces*. The antibiotics they produce can have biostatic and biocidal effects on soil-borne plant pathogens, including *Fusarium* which would have an incidence in these soils of less than 5 percent.



Autochthonous microorganisms

An indigenous form of soil microorganisms, responsible for chemical processes that occur in the soil under normal conditions.

Allochthonous microorganisms

Introduced microorganisms.

Biofertilizers



Biofertilizers are considered the alternate source of fertilizer to meet the nutrient requirement of crops and to bridge the future gaps.

Further knowing the deleterious effects of using only the chemical fertilizers, use of soil microorganisms, which can fix atmospheric nitrogen, solubilize phosphorus or stimulate plant growth through synthesis of growth promoting substances will be environmentally benign approach for nutrient management of ecosystem.



Biofertilizers are products containing living cells of different types of microorganisms, which have an ability to convert nutritionally important elements from unavailable to available form through biological processes

This group of organisms seems to play an important role in soil biofertility, because of their capability for producing hormones, amino acids and vitamins



In recent years, biofertilizers have emerged as an important component of the integrated nutrient supply system and hold a great promise to improve crop yields through environmentally better nutrient supplies.



A wide variety of biofertilizers is being used as a good source of plant nutrients in sustainable agriculture.

Microbes, especially some *Bacillus* strains (*B. polymyxa* as nitrogen fixer, *B. megaterium* as phosphate dissolving bacteria and *B. circulans* as potassium release bacteria) play central role in manufacturing of biofertilizers

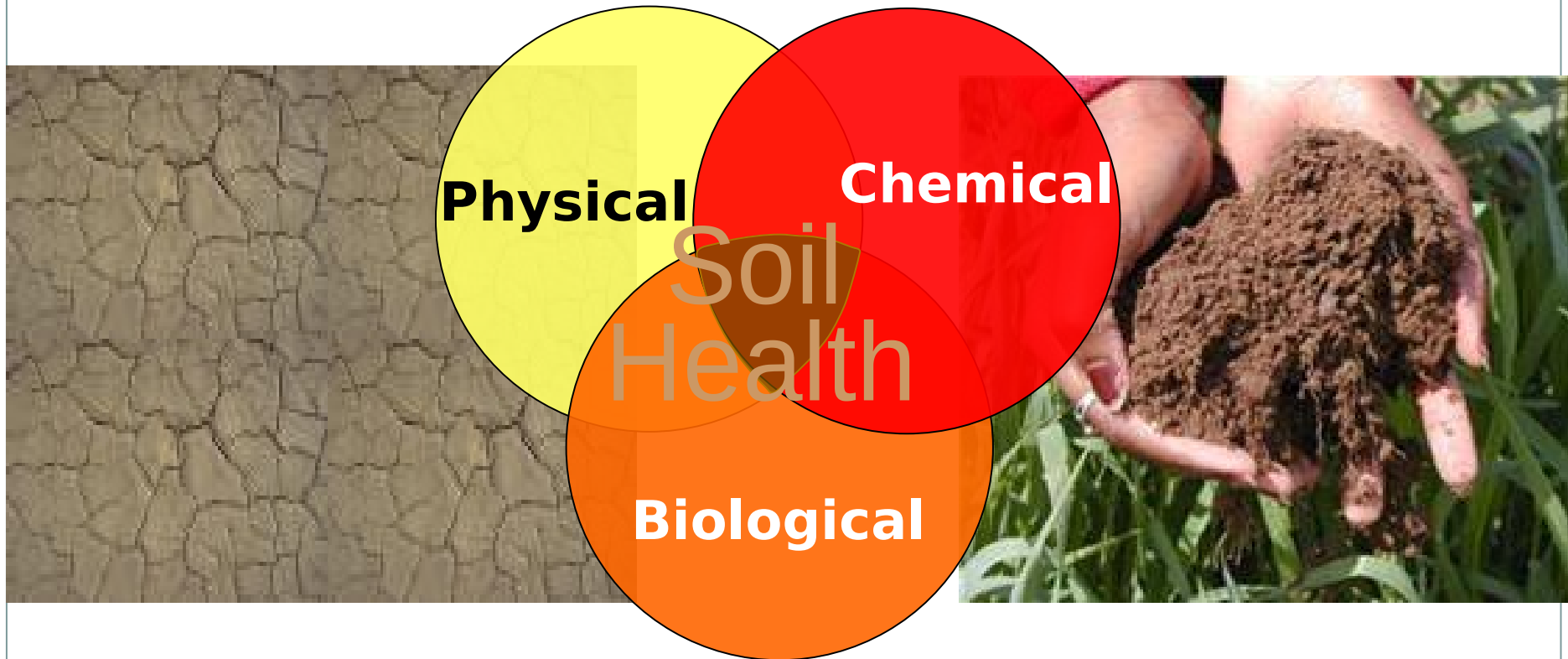


- The economic importance of biological NPK fertilizers and plant growth-promoting rhizobacteria (PGPR) inocula, not only felt in conserving the environment, but improve it, too.
- They reduced pollution hazards caused by leaching nitrate in the drainage water and through volatilization of NH_3 gas from $(\text{NH}_4)_2\text{SO}_4$ fertilizer and nitrogen oxide gases evolved during denitrification processes

Benefit of Biofertilizer

- To lessen the use of chemical fertilizer that may be a health hazard.
- Enhance nutrient uptake in deficient soils.
- They could secrete plant growth substances and antifungal chemical, and/or improve seed germination and root growth.

Soil Health Concept



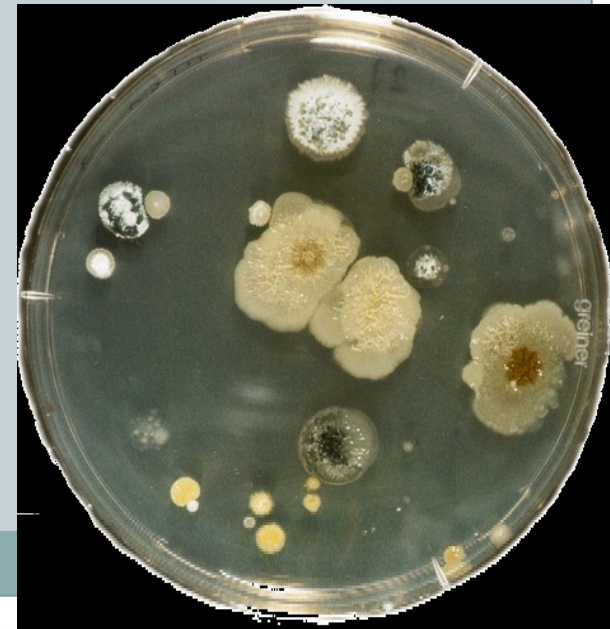
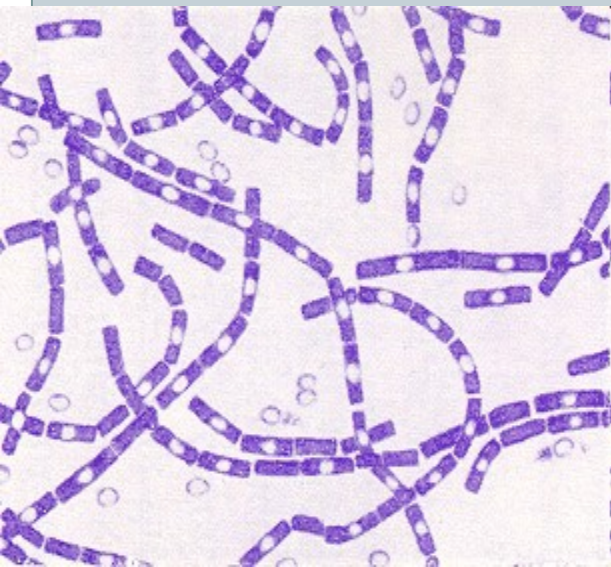
Emphasizing the integration of biological with chemical and physical measures of soil quality

as they affect grower profits and the

A healthy soil has....

- ✓ Good soil tilth
- ✓ Sufficient depth
- ✓ Sufficient but not excess supply of nutrients
- ✓ Good soil drainage
- ✓ Low population of plant pathogens and other pests
- ✓ High populations of beneficial organisms
- ✓ Low weed pressure
- ✓ Free of chemicals and toxins that may harm plants
- ✓ Resistant to degradation
- ✓ Resilient when unfavorable conditions occur

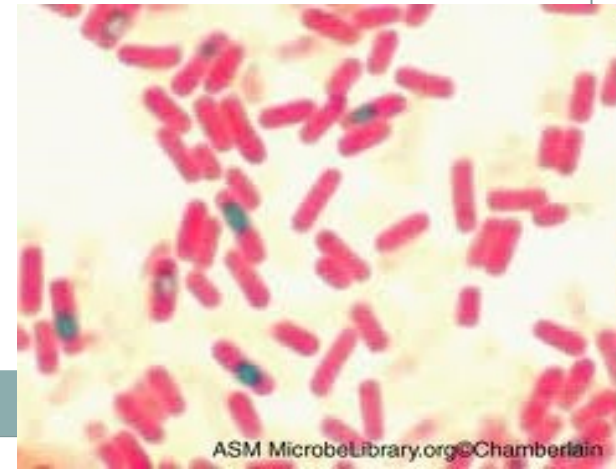
Examples of some Important soil microbes



Endospore-Forming Gram-Positive Bacteria



- Key Genera: *Bacillus*, *Clostridium*, *Sporosarcina*, *Heliobacterium*
 - Distinguished on the basis of cell morphology, shape and cellular position of endospore
 - Generally found in soils
 - Endospores are advantageous for soil microorganisms



Endospore-Forming Gram-Positive Bacteria



- *Bacillus and Paenibacillus*

- Many produce extracellular hydrolytic enzymes that break down polymers
- Many bacilli produce antibiotics
- *Paenibacillus popilliae* and *Bacillus thuringiensis* produce insect larvicides

Endospore-Forming Gram-Positive Bacteria



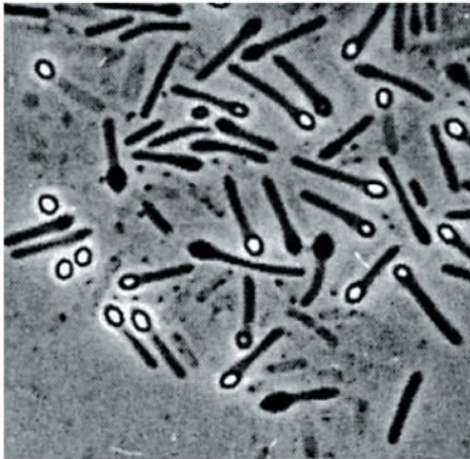
- *Clostridium*

- Lack a respiratory chain, anaerobic
- Some *Clostridia* perform **Stickland reactions**
 - ▮ Metabolism of pair of amino acids
- Mainly found in anaerobic pockets in the soil
 - ▮ Also live in mammalian intestinal tract
- Some are pathogenic; cause diseases such as **botulism**, **tetanus**, and **gas gangrene**

*Clostridium
botulinum*

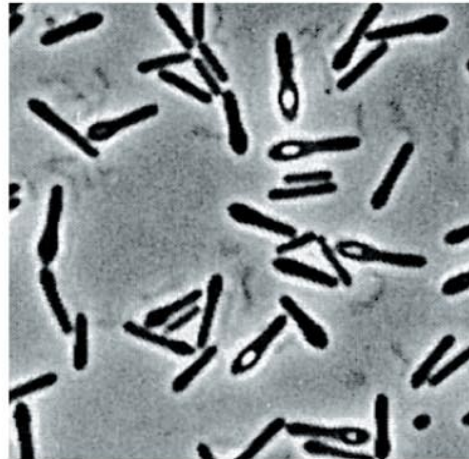


Clostridium Species and Endospore Location



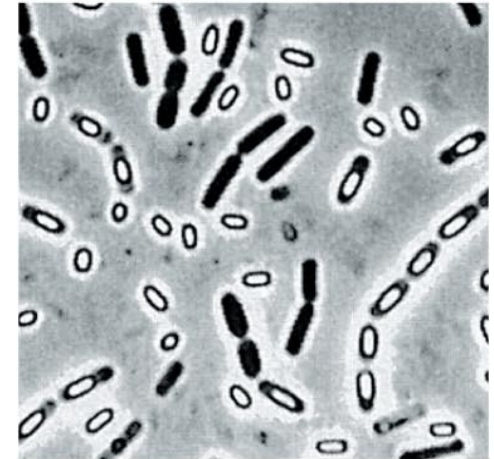
Hans Hippe

(a) *Clostridium cadaveris*
Terminal endospores



Hans Hippe

(b) *Clostridium sporogenes*
Subterminal endospores



Hans Hippe

(c) *Clostridium bifermentans*
Central endospores

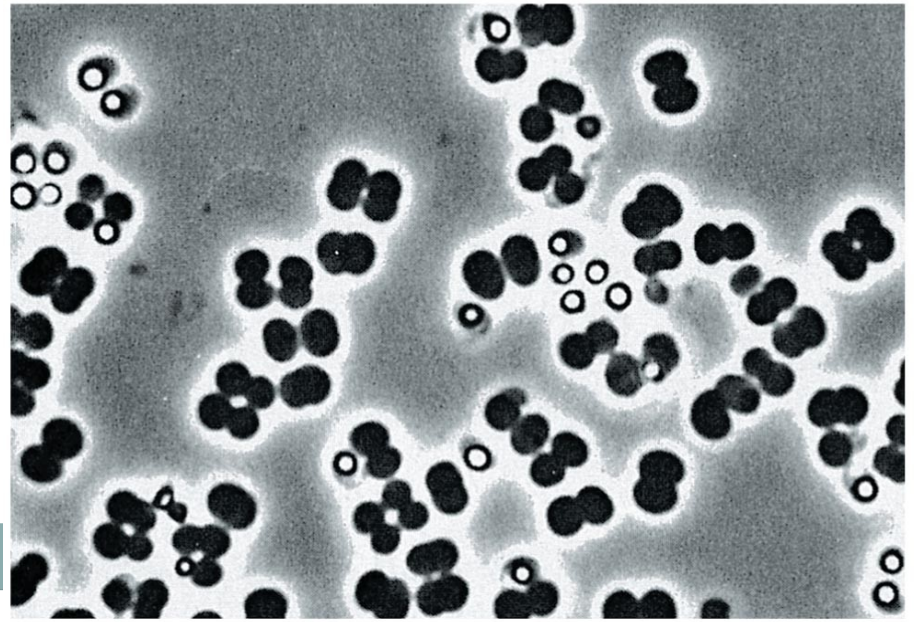
Endospore-Forming Gram-Positive Bacteria



- *Sporosarcina*

- Unique among endospore formers because cells are cocci instead of rods
- Strictly aerobic, spherical cells
- Common in soils

Sporosarcina ureae



Endospore-Forming Gram-Positive Bacteria



Heliobacteria

Phototrophic gram-positive bacteria

Anoxygenic phototrophs

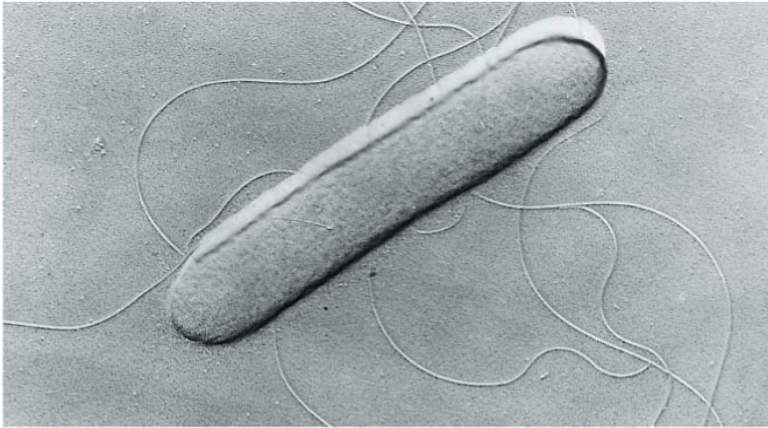
Produce bacteriochlorophyll *g*

Strict anaerobes

Reside in soils and also in highly alkaline environments (i.e.,
soda lakes and alkaline soils)

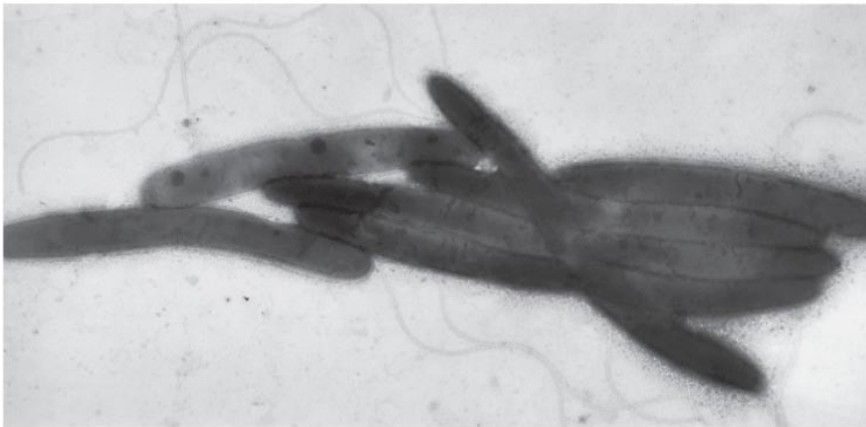
Have nitrogen-fixation capabilities

Cells and Endospores of Heliobacteria



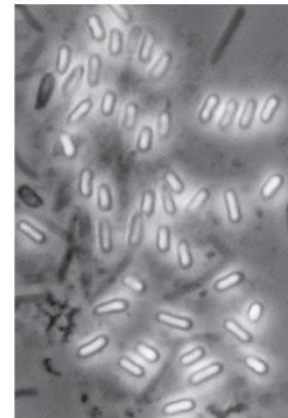
F. Rudy Turner & Howard Gest

(a) *Heliobacillus mobilis*



John Ormerod and M. T. Madigan

(b) *Heliophilum fasciatum*



John Ormerod and M. T. Madigan

(c) *Heliobacterium gestii*

Actinobacteria: Coryneform & Propionic Acid Bacteria



Key genera: *Corynebacterium*, *Arthrobacter*,
Propionibacterium

- Actinobacteria form their own phylum
 - High G+C gram-positive bacteria
- Over 30 taxonomic families
- Rod to filamentous, usually aerobic
- Mostly harmless commensals (*Mycobacterium* are notable exceptions)
- Valuable for antibiotics and certain fermented dairy products

Actinobacteria: Coryneform & Propionic Acid Bacteria



Corynebacterium

- Gram-positive, aerobic, non-motile, rod-shaped
- Form club-shaped, irregular-shaped, or V-shaped cell arrangements
- Extremely diverse

Arthrobacter

- Primarily soil organisms
- Remarkably resistant to dessication and starvation

Actinobacteria: Coryneform & Propionic Acid Bacteria



Propionic Acid Bacteria

- First discovered in Swiss cheese
- Gram-positive anaerobes
- Have metabolic strategy called **secondary fermentation**
 - ▮ Obtain energy from fermentation products produced by other bacteria
 - *Propionibacterium*: lactic acid
 - *Propionigenium*: succinate

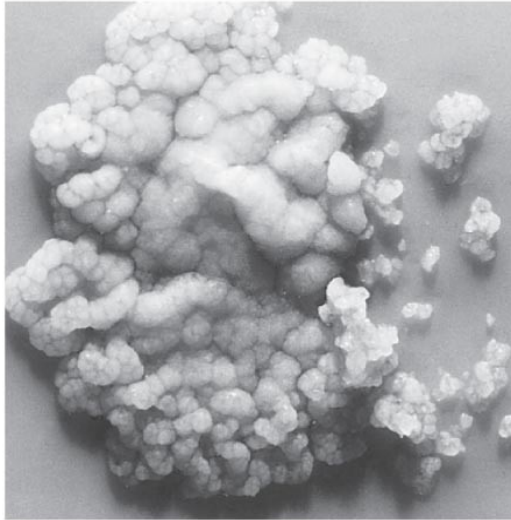
Actinobacteria: *Mycobacterium*



Mycobacterium

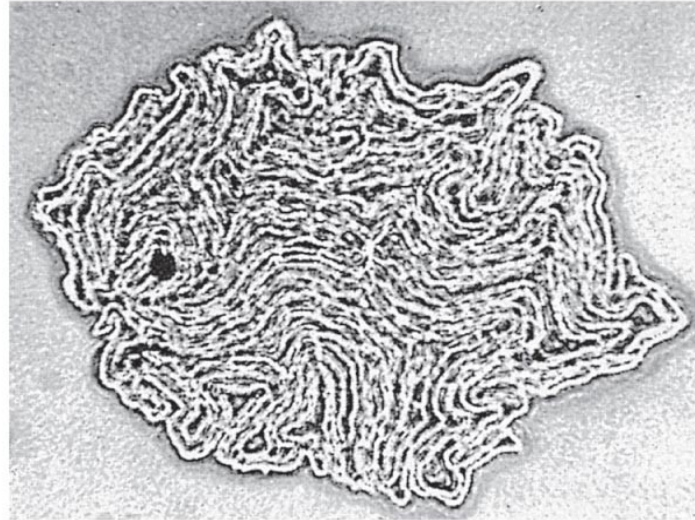
- Rod-shaped organisms, exhibit **acid-fastness**
- First discovered by Robert Koch
- Not readily stained by Gram stain because of high surface lipid content
- Cells are somewhat pleomorphic
- Separated into two groups: slow and fast growers
- Classified into three groups based on pigmentation

Characteristic Colony Morphology of Mycobacteria



N. Rist

(a) *M. tuberculosis* colony



V. Lorian

(b) Colony of virulent *M. tuberculosis* at an early stage



Centers for Disease Control

(c) *M. avium* colonies

Actinomycetes



These are soil organisms which have characteristics common to bacteria and fungi and yet possess sufficient distinctive features to delimit them into a distinct category. On agar plates, they can easily be distinguished from true bacteria.

Unlike slimy distinct colonies of true bacteria which grow quickly, actinomycete colonies appear slowly, show powdery consistency and stick firmly to agar surface.



Viruses are known to attack bacteria and actinomycetes but not fungi. Actinomycetes differ from fungi in the composition of their cell wall. They do not have chitin and cellulose which are commonly found in the cell walls of fungi.

Aerobic *Actinomyces*: Natural Habitats

- ***Nocardia* species and other aerobic *Actinomyces* ubiquitous in soil and primarily responsible for decomposition of organic plant matter**
- ***Rhodococcus* species present in the intestinal bacterial flora of grazing herbivores especially horses**
- ***Streptomyces* species (>3,000) widely distributed in soil**

Taxonomy of the Aerobic *Actinomycetes*

Gram-positive branching filaments that sporulate or fragment: the aerobic *Actinomycetes* (order *Actinomycetales*)

Aerobic *Actinomycetes* whose cell walls contain mycolic acid: *Nocardia* species and *Rhodococcus* species (family *Nocardiaceae*)

Aerobic *Actinomycetes* whose cell walls lack mycolic acid: *Streptomyces* species



The number of actinomycetes increases in the presence of decomposing organic matter. As a rule, they are intolerant to acidity and their numbers decline at pH 5.0. The most conducive range of pH is between 6.5 and 8.0.

Waterlogging of soil is unfavourable for the growth of actinomycetes whereas desertic soils of arid and semi-arid zones sustain sizeable population, probably due to the resistance of spores to desiccation.



The percentage of actinomycetes in the total microbial population increases with the depth of soil and actinomycetes can be isolated in sufficient number even from soil samples obtained from the C horizon of a soil profile.

Filamentous Actinobacteria: *Streptomyces* & Others



- Key genera: *Streptomyces*, *Actinomyces*
 - Filamentous, gram-positive bacteria
 - Produce **mycelium** analogous to mycelium of filamentous fungi
 - Over 500 species of *Streptomyces* are recognized
 - *Streptomyces* spores are called **conidia** (in the aerial mycelium **sporophores**)
 - Primarily soil microorganisms, responsible for earthy odor of soil (**geosmins**)
 - aerobic and produce many extracellular enzymes. *Actinomyces* are anaerobic.

Filamentous Actinobacteria: *Streptomyces* & Others

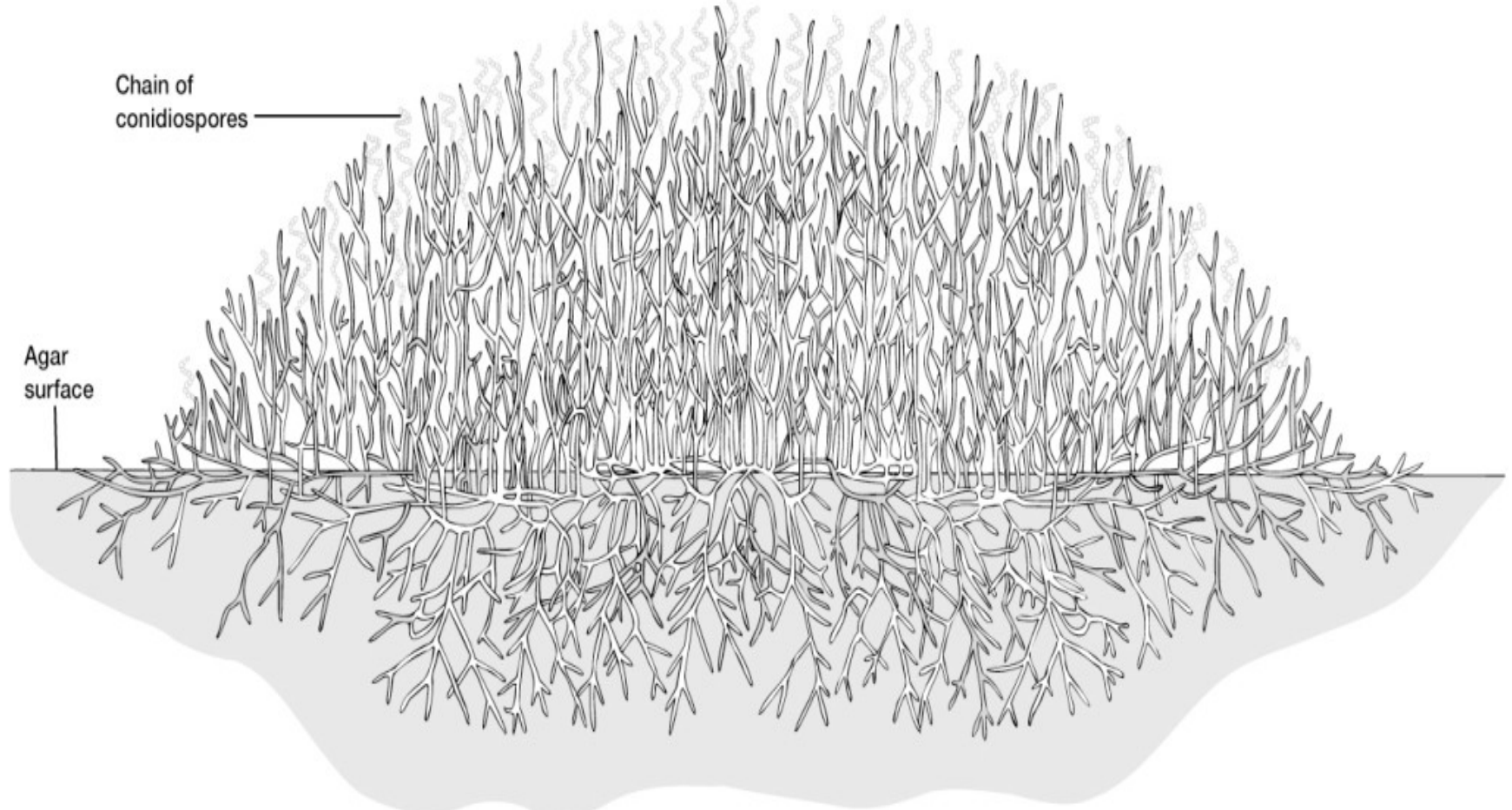


- ***Streptomyces***

- 50% of all isolated *Streptomyces* produce **antibiotics**
- Over 500 distinct antibiotics produced by *Streptomyces*
- Some produce more than one antibiotic
- Knowledge of the ecology of *Streptomyces* remains poor

ACTINOMYCETES MORPHOLOGY

Copyright © The McGraw-Hill Companies, Inc. Permission required for reproduction or display.

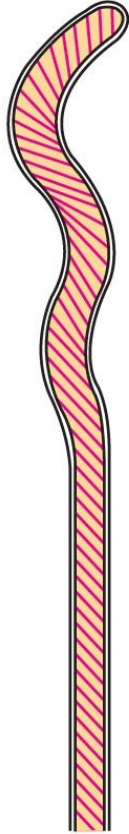


An actinomycete colony growing on agar - note the subterranean and aerial hyphae....

Spore Formation in *Streptomyces*



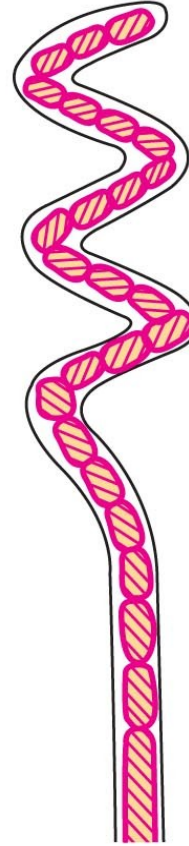
**Growth
phase**



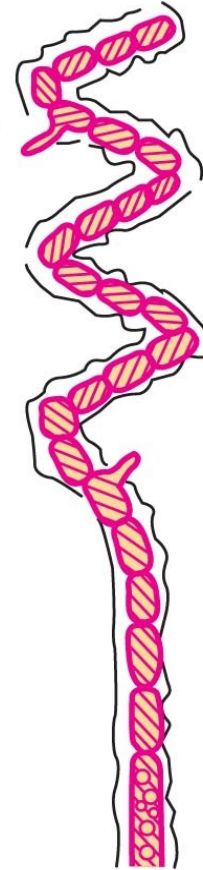
**Tip
curls**



**Partitioning
of tip**



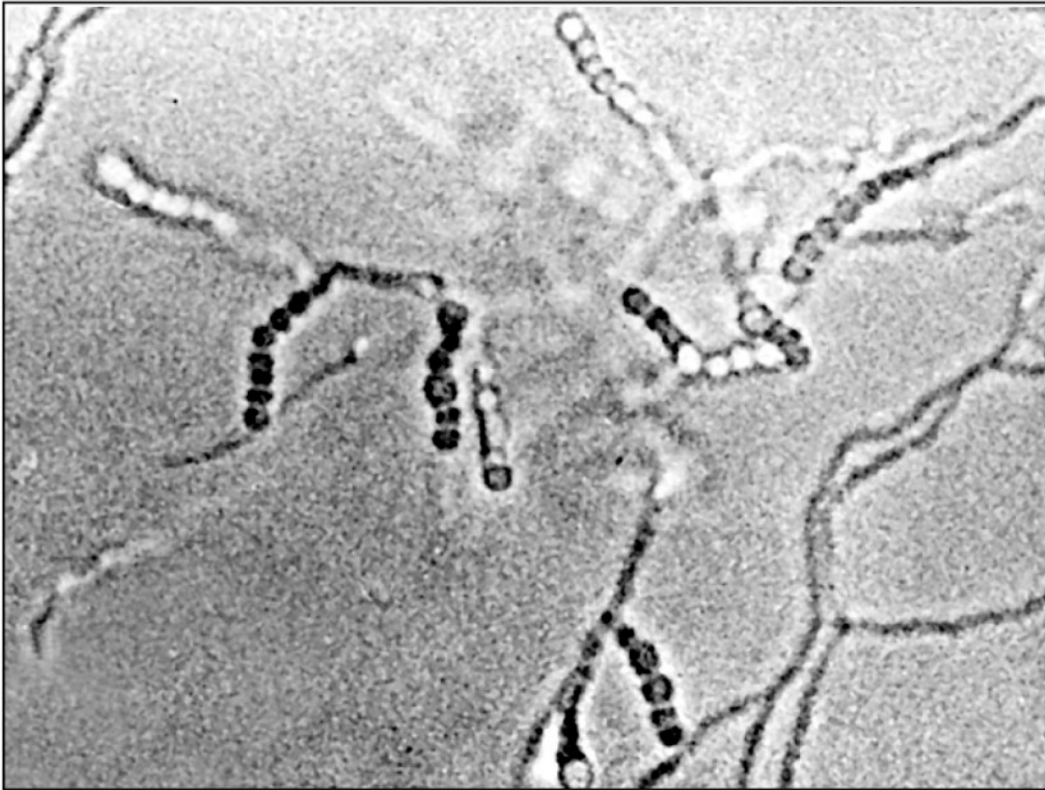
**Cell walls
thicken
and constrict**



**Spores
mature**

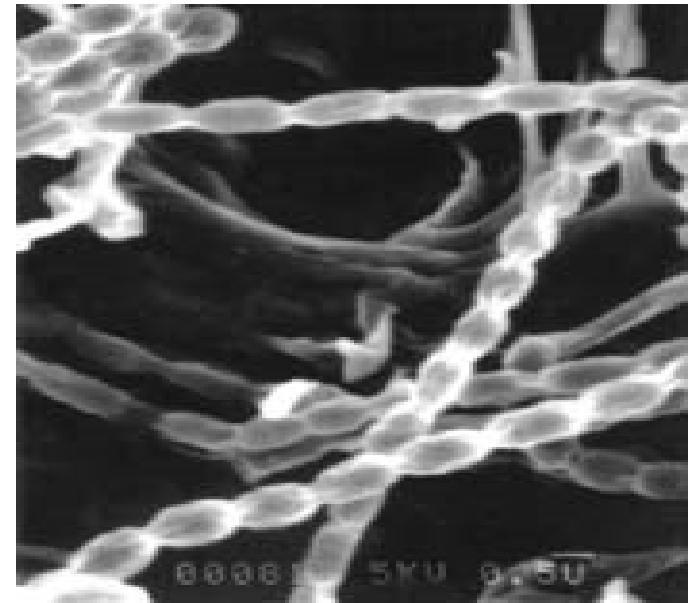
ACTINOMYCETES REPRODUCTION

Copyright © The McGraw-Hill Companies, Inc. Permission required for reproduction or display.



Conidia of *Streptomyces*

5 μ m



High Magnification of
actinomycete spores

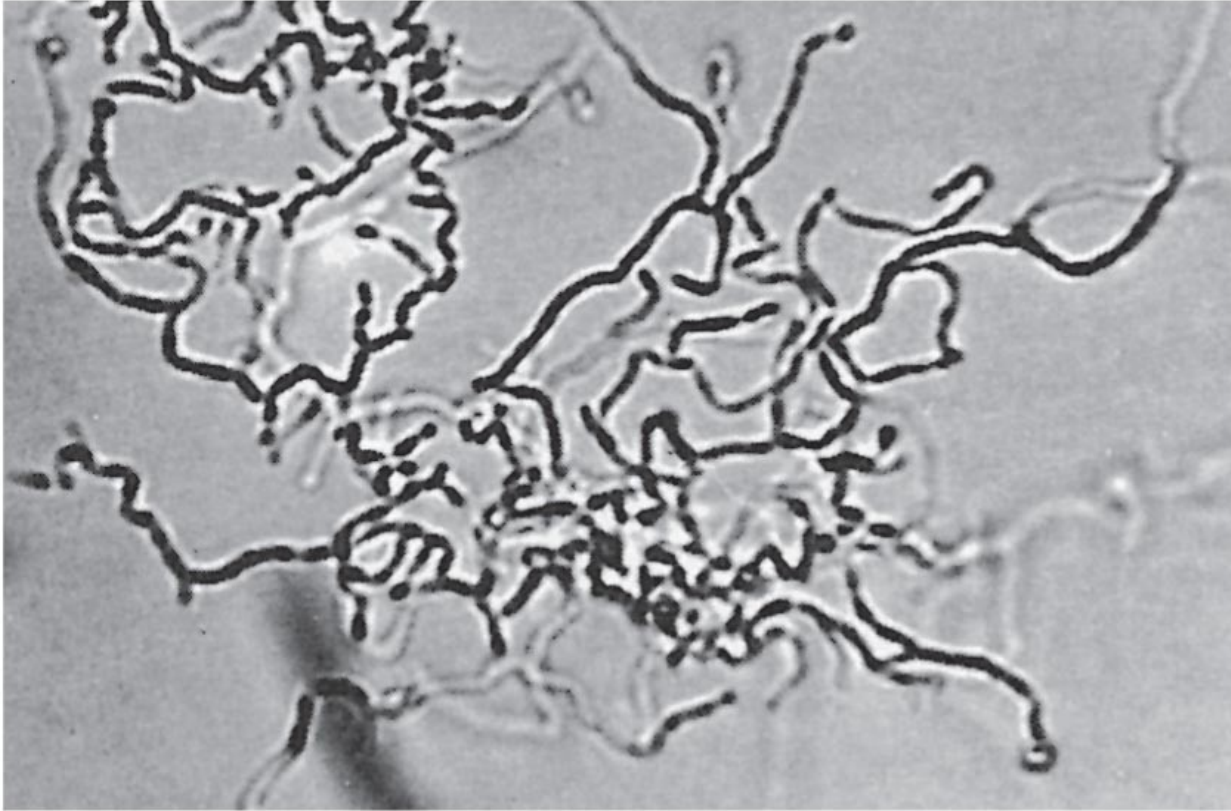
ACTINOMYCETES REPRODUCTION

Actinomycetes (and fungi) reproduce via spores
hyphal growth is followed by fragmentation
and release of spores

conidia – spores produced asexually on aerial
filaments called sporophores

shape and arrangement of sporophores help
identification of actinomycete taxa

Nocardia



Hubert and Mary P. Lechevalier

Figure 16.18

Spore-Bearing Structures of Actinomycetes



Peter Hirsch

(a) *Streptomyces*
sp.

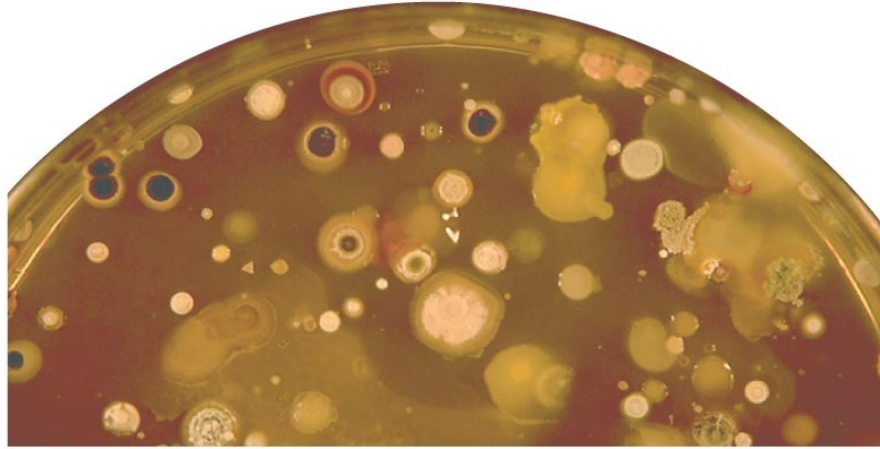


Hubert and Mary P. Lechevallier

(b) *Streptomyces*
sp.

Figure 16.19

Colonies of *Streptomyces*



M. T. Madigan

(a)



David A. Hopwood

(b)

Figure 16.22

ACTINOMYCETES ECOLOGY



predominantly soil bacteria
good at degrading recalcitrant compounds
such as chitin & cellulose
often active at higher pH (contrast to fungi who
may dominate at lower pH)
give soil the “earthy” smell - geosmin

Examples



Genus *Frankia* –

large genus that all form nodules on certain non-legume trees

fixes nitrogen from the air and converts into forms useable by plant host

involved in relationship with plant and mycorrhizal fungus. e.g.

Myrica in the tropics (invasive in Hawaii)

Alnus in temperate climates (early successional)

Frankia nodules on *Ceanothus* roots



Other antibiotics produced by *Streptomyces* spp

Antibiotic:

Spectinomycin
gonorrhoeae

affects:

M. tuberculosis, Neisseria

Neomycin

Broad spectrum,

Tetracyclines

Broad spectrum, Chlamydias,
Rickettsias etc.

Nystatin

Fungi, esp. *Candida*

Erythromycin

Gram + Bacteria and *Legionella*

Chloramphenicol

Broad spectrum, typhoid fever

Recommended Reading-General Microbiology



- GJ Tortora, Christine L and Funke BR (2007). Microbiology: an introduction, 9th ed. Pearson Education, New Delhi.